Institute of Mathematical Sciences
University of Malaya

SIM1001 Basic Mathematics

SIX1013 Fundamentals of Advanced Mathematics

TUTORIAL 1 (Part 1)

1. Which of the following sentences are statements?
   If it is a statement, determine its truth value.

   (i) -1 or 0 is a positive number.

   (ii) Do not sleep here.

   (iii) A rectangle has four sides but a square has five sides.

   (iv) Chocolate cakes are tastier than cheesecakes.

   (v) Neither 4 nor 7 is a factor of 47.

   (vi) $1 - 6 = 6 - 1$?

   (vii) The real number x is rational.

   (viii) There exists a real number x such that x is rational.

2. (a) Write each of the following statements in symbolic form using statement variables and connectives;

   (b) Write the negation of each of the following statements in both symbolic form and in English.

   (i) The main campus of University of Malaya is in Kuala Lumpur and Kuala Lumpur is the capital of Malaysia.

   (ii) Neither $\sqrt{2}$ nor π is an integer.

   (iii) John likes to play football and badminton but he does not like to swim.

   (iv) The bus was late or John's watch was slow.

   (v) A fish can walk or it has no leg.

   (vi) 5 or 8 is a multiple of 2, but the sum of 5 and 8 is not multiple of 2

3. Construct the truth table for each of propositions in (i) – (v).

   (i) $\sim(p \vee \sim q) \wedge q$ (ii) $\sim(p \wedge \sim r) \vee (q \wedge r)$

   (iiii) $(p \wedge \sim q) \vee (p \wedge r)$ (iv) $p \rightarrow q \wedge \sim r$

   (v) $(p \rightarrow \sim q) \leftrightarrow \sim (q \vee r)$

   Which of the proposition in (i) – (v) is a tautology and which is a contradiction?

4. By using truth tables, determine which of the pairs of propositions in (i) – (v) are logically equivalent.

   (i) $\sim p \wedge \sim (p \wedge q)$ and $p \vee (\sim q \wedge \sim p)$

   (ii) $(r \vee p) \wedge \{[\sim r \vee (p \wedge q)] \wedge (r \vee q)\}$ and $(p \wedge q) \vee (r \wedge \sim r)$

   (iii) $p \vee \sim q$ and $\sim (p \wedge q) \rightarrow \sim q$

   (iv) $p \vee q \rightarrow r$ and $(p \rightarrow q) \wedge (q \rightarrow r)$

   (v) $(q \rightarrow p) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$

   (vi) $(p \rightarrow q) \leftrightarrow r$ and $(p \rightarrow r) \leftrightarrow (q \rightarrow r)$

5. Use the laws of logical equivalences, simplify each of propositions in (i) – (v).

   (i) $p \wedge [\sim (\sim p \vee q) \vee (p \wedge q)]$

   (ii) $(p \wedge q) \vee [(r \wedge q) \vee (r \wedge \sim q)]$

   (iii) $[(p \wedge q) \vee (q \wedge r)] \vee [(p \wedge r) \vee r]$

   (iv) $p \vee \{\sim p \rightarrow [q \vee (\sim q \rightarrow r)]\}$

   (v) $[\sim (p \rightarrow q) \vee p] \wedge [(p \wedge \sim q) \vee q]$

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